

Week 10 – Performance Optimization & Concurrency

Summary:

Week 10 focused on improving **execution speed and concurrency** for multiple simultaneous requests.

Key Activities:

- Optimized Python and system calls for faster compilation and execution.
- Implemented **asynchronous execution** for concurrent multi-language code submissions.
- Monitored **CPU and memory usage** to prevent overloads.
- Tested multiple simultaneous requests in Docker to verify performance.

Technologies Used: Python, FastAPI (async), Docker, WSL2, GCC, g++, JDK

Outcome:

Backend is optimized for **faster execution**, can handle **multiple simultaneous code submissions**, and remains stable under load.